CELEBAL EXCELLENCE INTERNSHIP PROGRAM

DATA ENGINEERING

WEEK - 4 ASSIGNMENT REPORT

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

SUBMITTED BY:-

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(JKLU)

Objective:

The objective of this assignment is to understand Azure cloud services and build a complete data pipeline using Azure Storage Account and Azure Data Factory. The pipeline reads a CSV file from Blob Storage, validates it using metadata checks, and copies it to a destination location.

➔ Task 1: Resource Group Creation

A Resource Group named “**Week4-Assignment**” was created in the Azure Portal to manage all resources used in this assignment. It serves as a logical container for organizing Azure services.

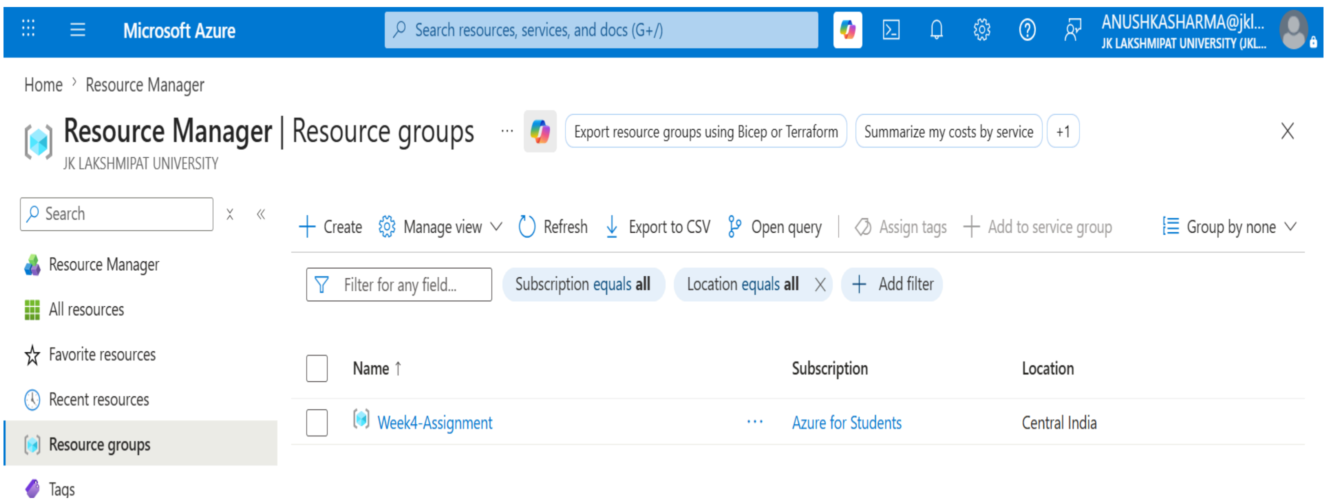

Figure 1: Resource Group : Week4-Assignment

➔ Task 2: Storage Setup

A Storage Account named **“w4assignstorage”** was created to store the data used in this assignment. A Blob container named **“source-data”** was created, and the dataset **“Sample - Superstore.csv”** was uploaded to it for use as the source file in the Azure Data Factory pipeline.

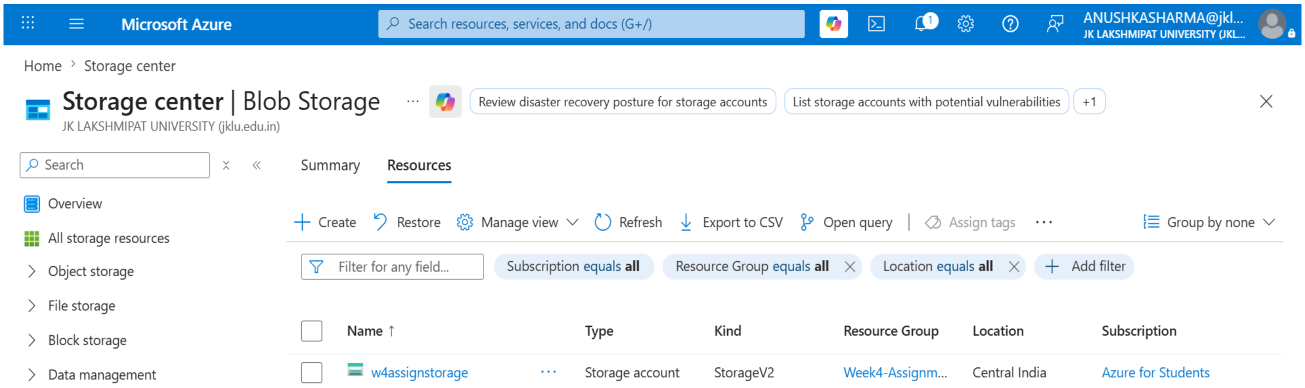

Figure 2: Storage Account : w4assignstorage

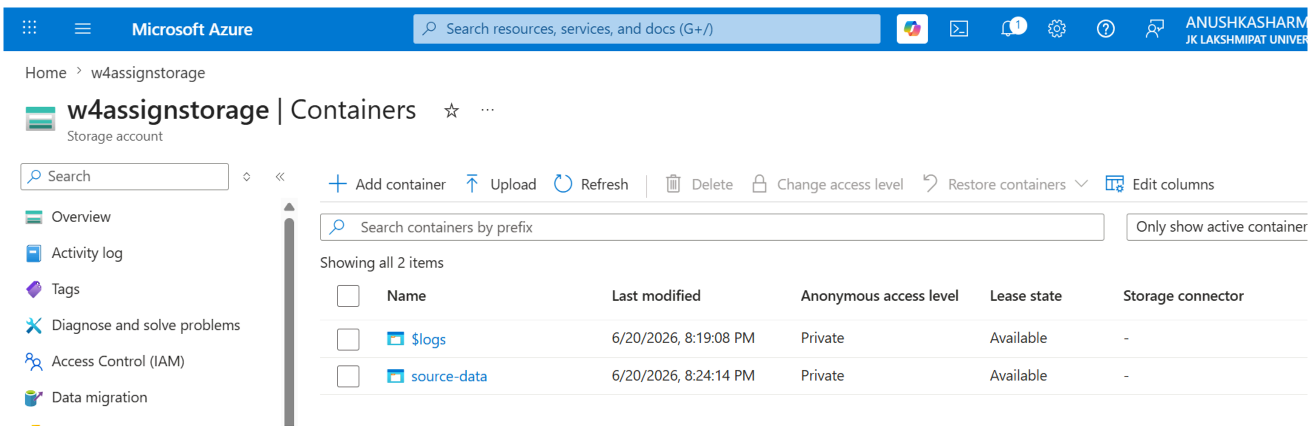

Figure 3: Blob Container : source-data

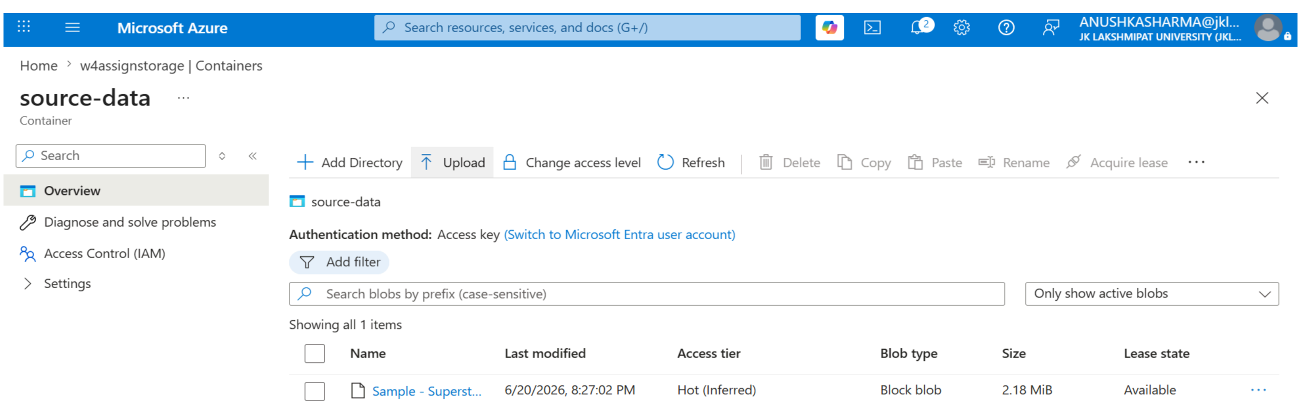

Figure 4: Uploaded File : Sample - Superstore.csv

➔ Task 3: Azure Data Factory Basics

Azure Data Factory Creation

An Azure Data Factory instance named **“ADF-W4Assign”** was created and the Author, Monitor, and Manage sections were explored. A Linked Service named **“AzureBlobStorage1”** was configured to connect ADF with Blob Storage.

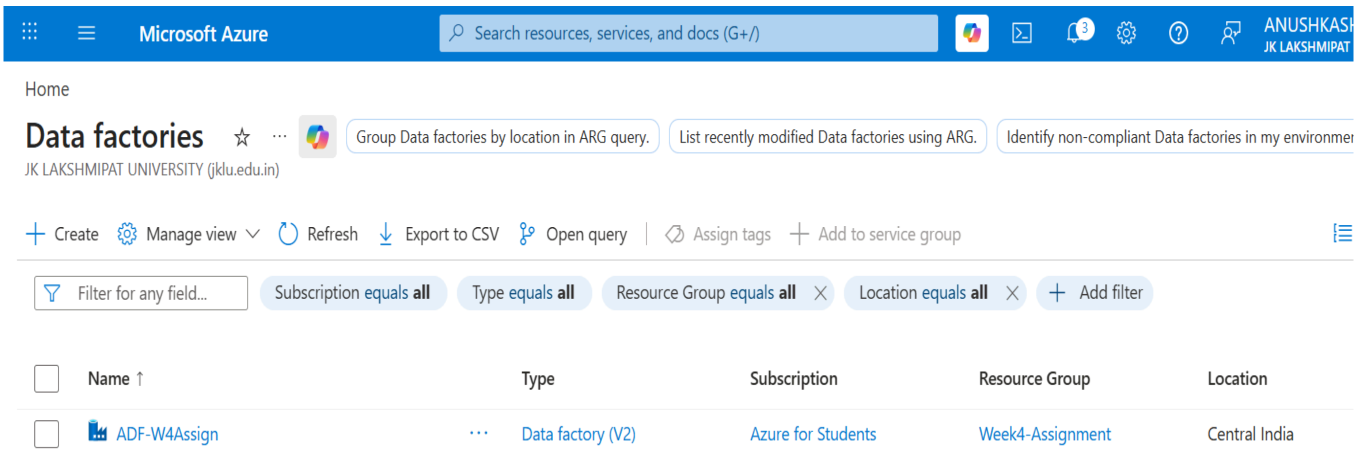

Figure 5: Azure Data Factory : ADF-W4Assign

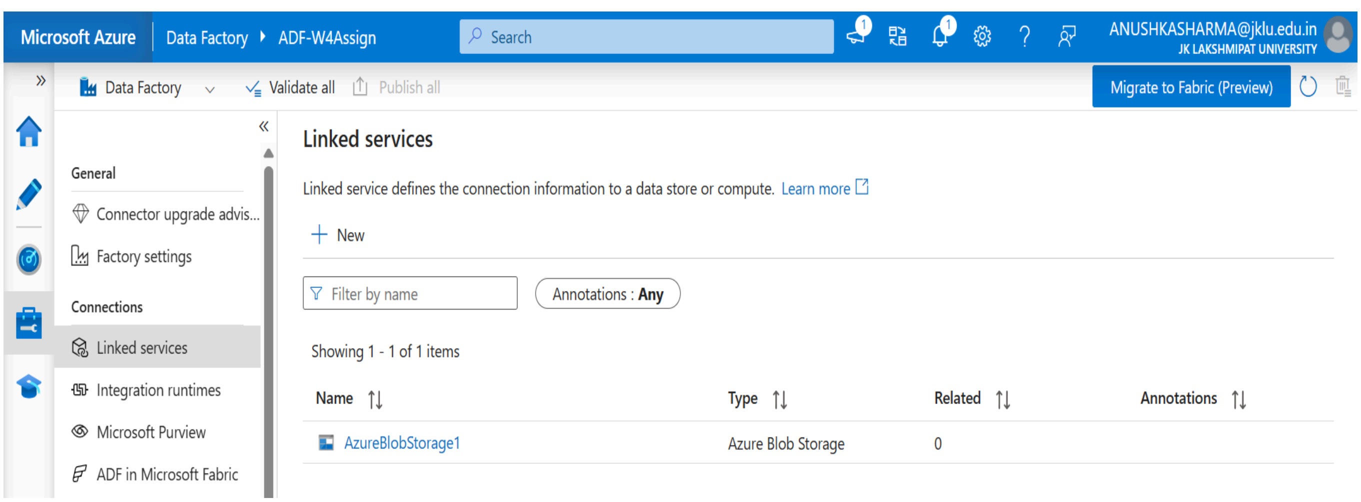

Figure 6: Linked Service : AzureBlobStorage1

Dataset Creation

Two datasets were created:

- **Superstoredata** (Source Dataset)
- **ProcessedData** (Destination Dataset)

These datasets were used to define the source and destination files for the pipeline.

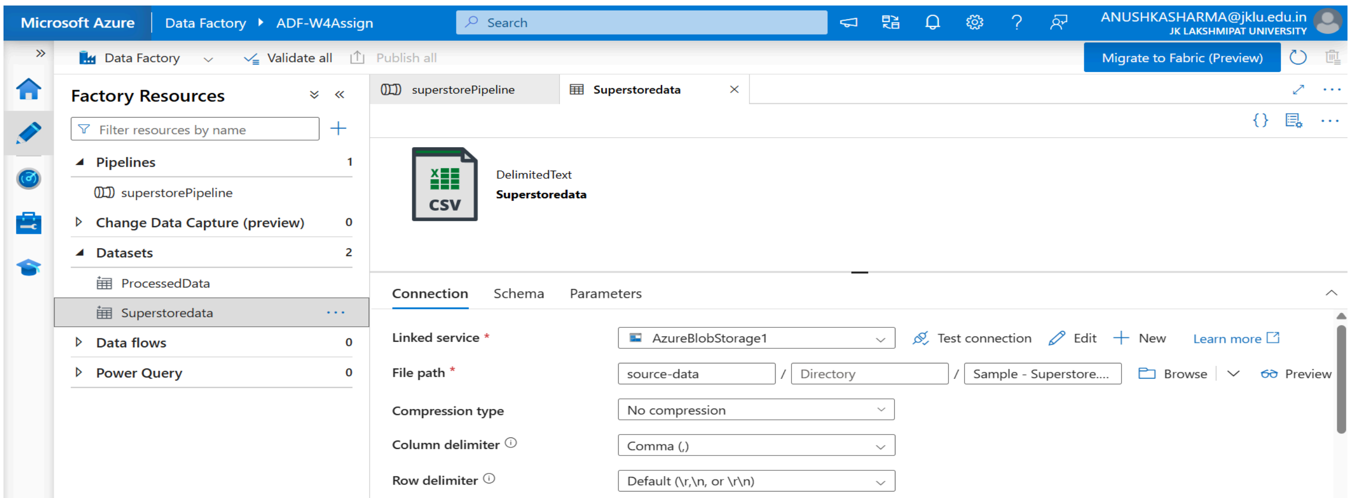

Figure 7: Source Dataset : Superstoredata

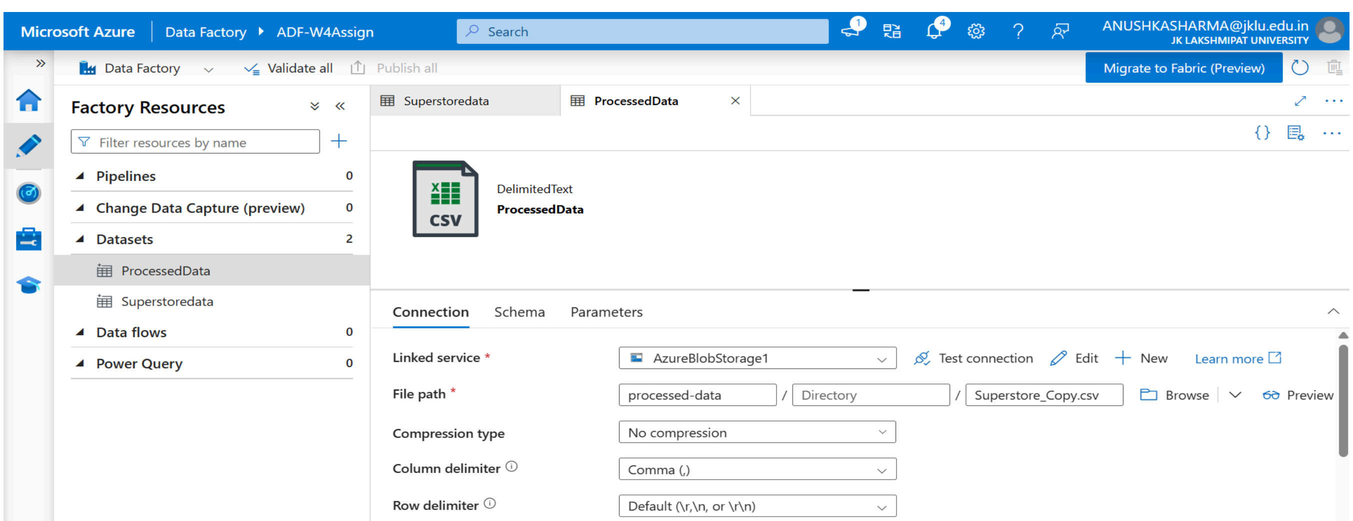

Figure 8: Destination Dataset : ProcessedData

Get Metadata Activity

A Get Metadata activity was added to validate the source file before processing. The activity was configured to check the file existence and size.

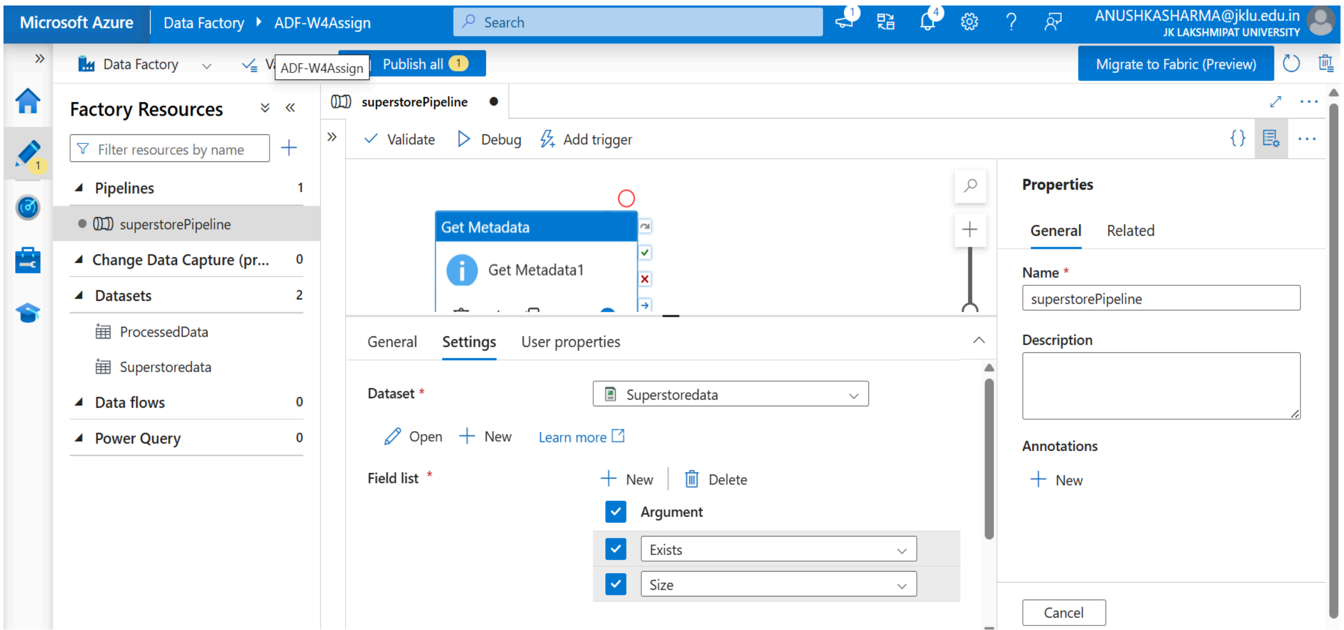

Figure 9: Get Metadata Activity Configuration

➔ Task 4: Pipeline Development

A pipeline named **"superstorePipeline"** was created using Get Metadata and Copy Data activities. The pipeline reads the source file from Blob Storage, validates it using Get Metadata, and then copies it to the destination container. Since only a single file was processed, the ForEach activity was not required.

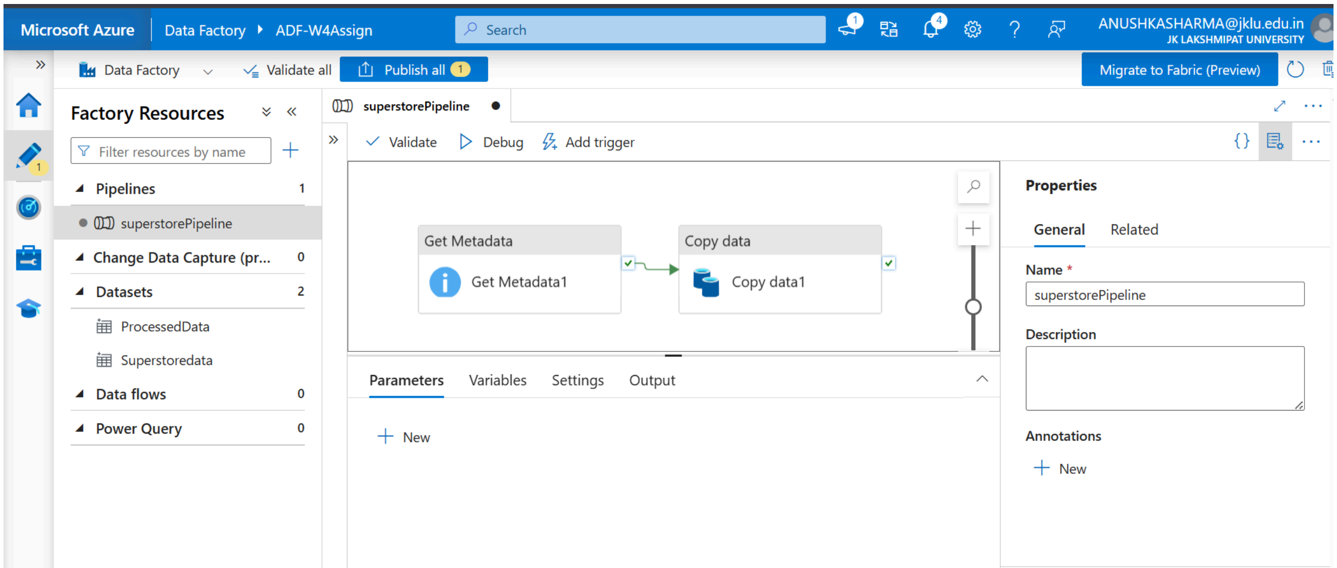

Figure 10: Pipeline Design : superstorePipeline

➔ Task 5: Pipeline Execution

The pipeline was executed using the Debug option in Azure Data Factory. Both Get Metadata and Copy Data activities completed successfully, resulting in a successful pipeline run.

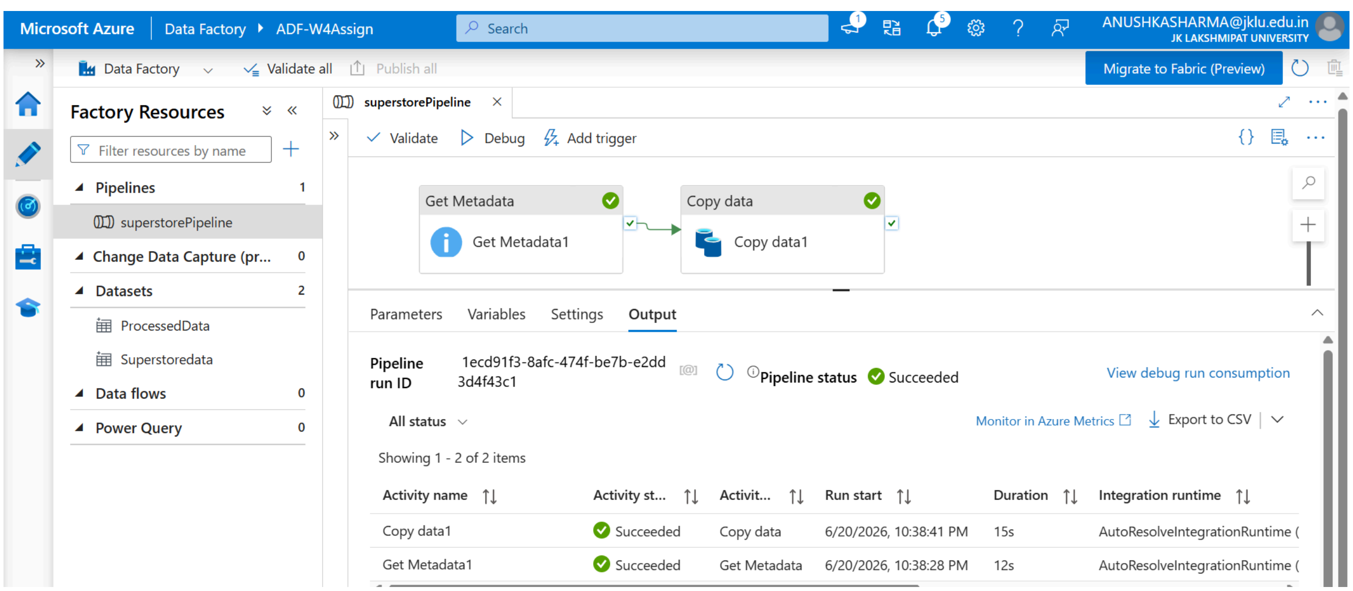

Figure 11: Successful Pipeline Execution

➔ Task 6: IAM Role Assignment

The **Reader** and **Contributor** roles were assigned. Additionally, **Storage Blob Data Contributor** access was provided to **ADF-W4Assign** so that Azure Data Factory could access the Storage Account.

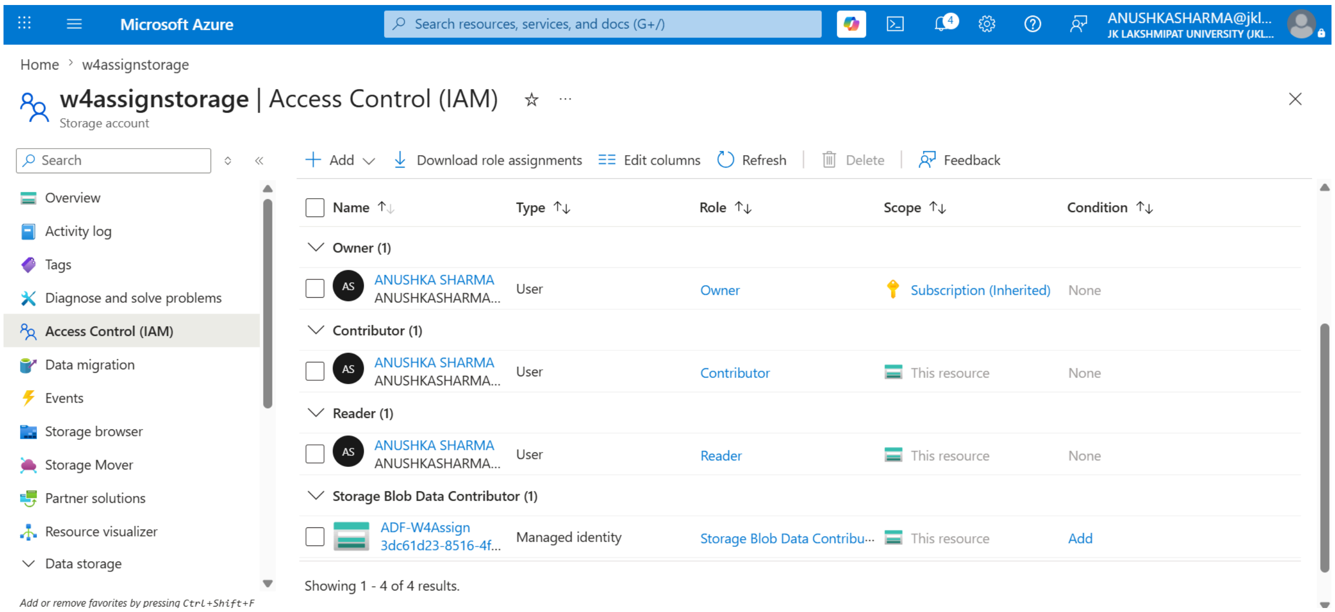

Figure 12: IAM Role Assignments and ADF Storage Access

➔ Mini Project: End-to-End Data Pipeline

A complete data pipeline was built using Azure Data Factory to process a CSV file stored in Azure Blob Storage. The pipeline validates the file using Get Metadata and then copies it to a new location using the Copy Data activity.

Components Used :

- Resource Group: **Week4-Assignment**
- Storage Account: **w4assignstorage**
- Azure Data Factory: **ADF-W4Assign**
- Linked Service: **AzureBlobStorage1**

- Pipeline: **superstorePipeline**
- Source Dataset: **Superstoredata**
- Destination Dataset: **ProcessedData**

Output : The pipeline executed successfully and generated **"Superstore_Copy.csv"** inside the **"processed-data"** container. Metadata validation was completed successfully before the file was copied.

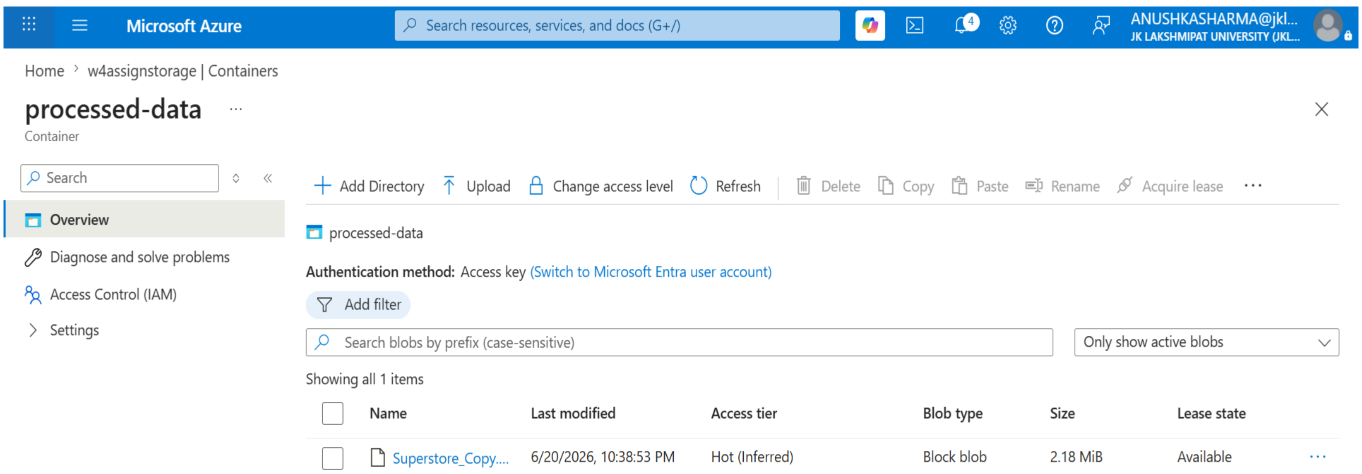

Figure 13: Superstore_Copy.csv in processed-data Container

➔ Conclusion

This assignment provided hands-on experience with Azure Storage, IAM roles, and Azure Data Factory. A complete pipeline was successfully created to validate and copy data from a source Blob container to a destination location using Azure Data Factory.